

WASP-163b

R-2825

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_260801 · created 2026-09-05T22:53:19+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461253.6832 +/- 0.0077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1488 +/- 0.008
Transit depth (Rp/Rs)^2	2.21 +/- 0.24 [%]
Orbital Inclination (inc)	82.19 +/- 2.48
Airmass coefficient 1 (a1)	1.197 +/- 0.014
Airmass coefficient 2 (a2)	-0.1128 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	11.63
Optimal Annulus	22.29
Transit Duration (day)	0.065 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1488 ± 0.008 vs 0.119 (deviation 2.34σ)
Duration fit vs published	0.065 d vs 0.0942 d (deviation 0.83σ)
Mid-time O–C	-77.6 min
Depth SNR	11.7
Residual scatter	1.13 %

Light curve

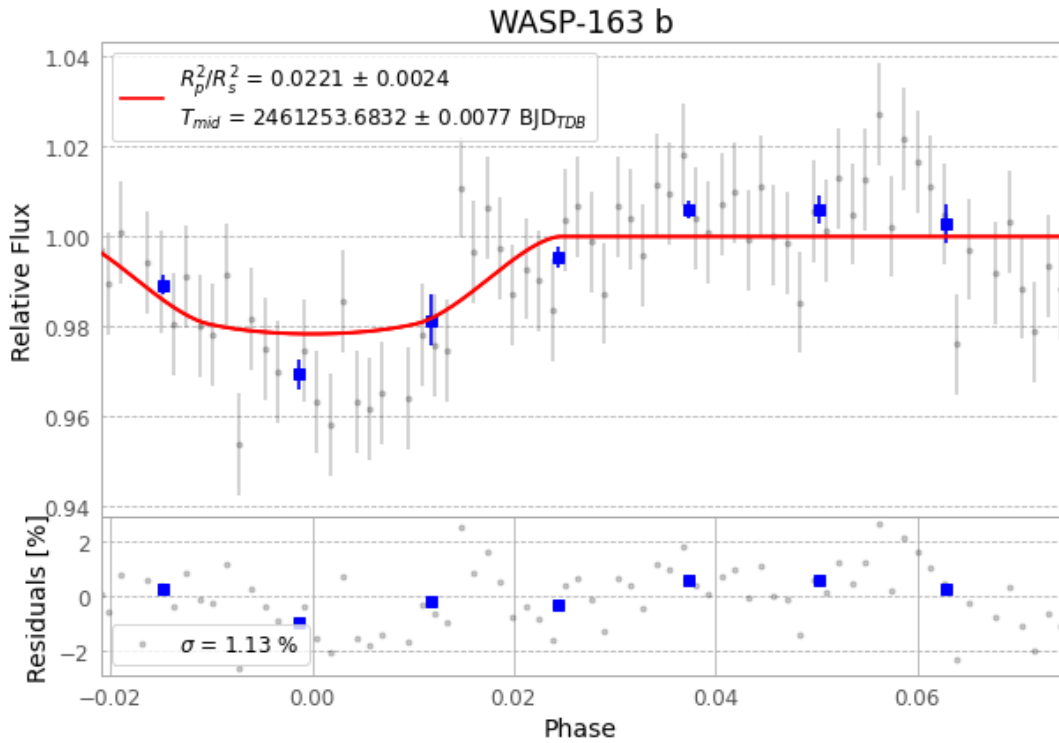


Figure 1 — FinalLightCurve_WASP-163 b_2026-07-31.png (SHA-256 d5485122cb5a8dcc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [605, 342], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1e609c312f7762b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), WASP-163260801032833.FITS → WASP-163260801070916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260801.log (a894bd385bc2...), FinalLightCurve_WASP-163 b_2026-07-31.png (d5485122cb5a...), FinalLightCurve_WASP-163 b_2026-07-31.csv (aeb5c5a5a170...)

Compute resources

Wall time 191.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 22:53Z.